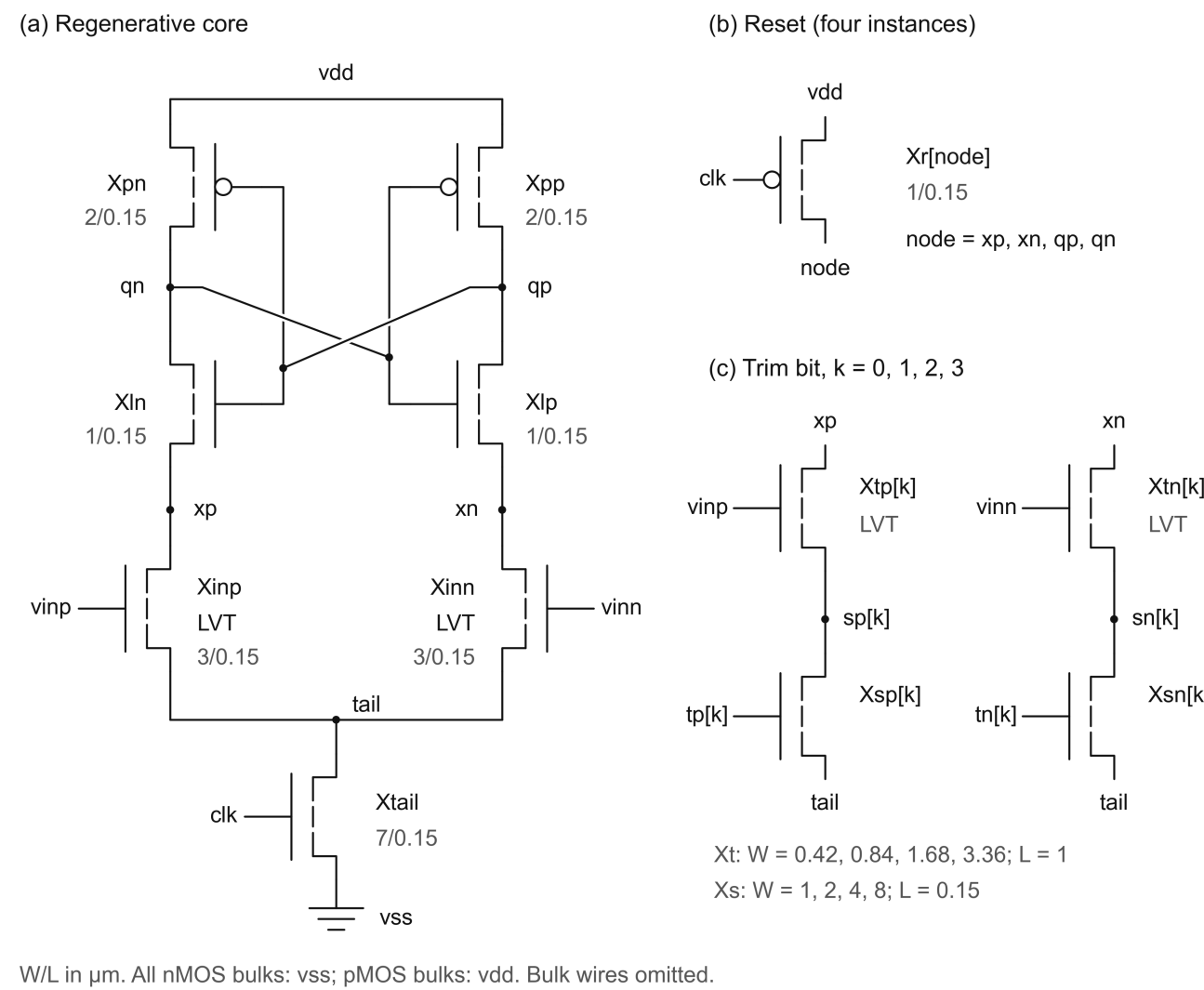


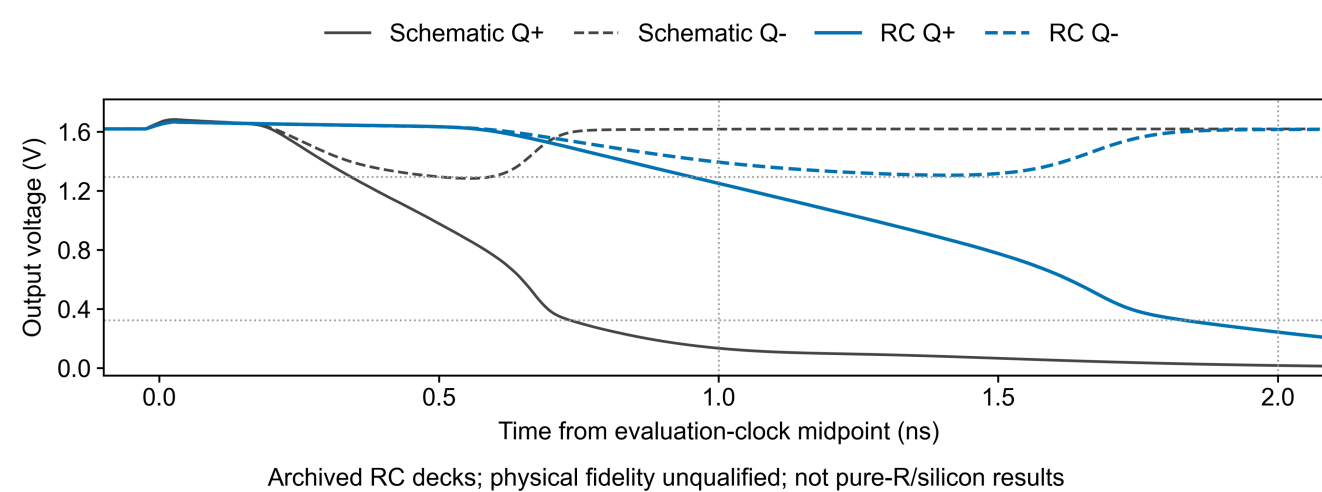
1 | Circuit and evaluation

27-device SKY130 StrongARM; LVT inputs, SVT latch.
Source-gated trim: sign plus four magnitude bits.



Nine declared candidates; separate lower-energy control.
Local calibration, finite decision deadline, complementary
80% / 20% output rails. No universal best-design claim.

4 | Retained SC / RC waveform



FS / 1.62 V / -40 C / -3 mV. Nominal, code zero; 5 fF per
output. The archived RC trace misses 1 ns and reaches the
required rails at 1.835 ns.

[Comparator_Atlas.ipynb](#)

[Colab](#)

Archived RC decks; physical fidelity unqualified; not pure-R/silicon results. Core energy excludes external drivers and calibration infrastructure.

Established circuitry: Razavi, 2015, DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka, 2022, DOI 10.1007/s10470-022-01992-6.

Original code: MIT. SKY130 keeps Apache-2.0. GitHub Copilot assistance disclosed. Source, models and reproduction instructions are linked in the notebook.

2 | Calibrated schematic decisions at 1 ns

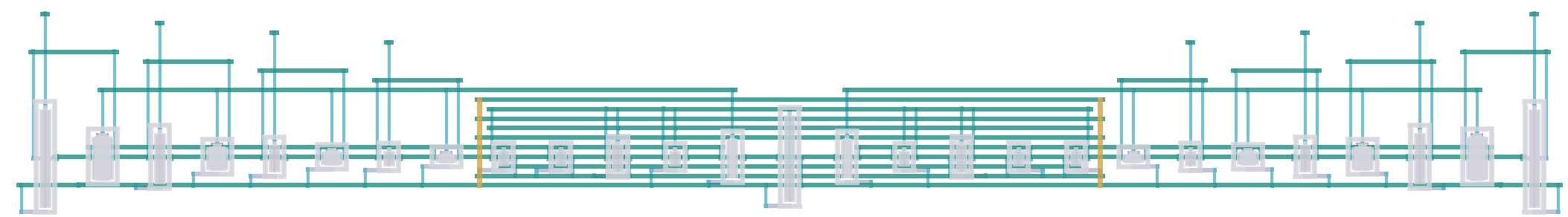
Same local calibration; 1 ns; sampled |input| ≥ 1 mV; +4% branch-width stress



All 49 conditions: 79.1% -> 94.9% correct | 143.0 -> 251.5 fJ/cycle

Above: 45 PVT combinations; aggregate also includes four nominal stress controls. Percentages are grid coverage, not yield.

3 | Layout checks and nominal RC results



27-device GDS | DRC 0 + LVS / negative controls | 2207.088 μm^2

Archived 45-PVT RC: 156/180 at 1 ns | 180/180 at declared 2 ns

Worst sampled RC decision: 1.835 ns at FS / 1.62 V / -40 C / -3 mV. Four signed inputs per condition; 24 unresolved at 1 ns. Separate ngspice-42 pilot: 12/20 at failed original 1 ns; 20/20 at post-hoc 2 ns. No continuous-input or statistical-yield guarantee.

Matched 180-point mean core energy, same ngspice-47 conditions:

Schematic: 245.8 fJ | Extracted RC: 425.5 fJ